

Global Equity Derivatives & Strategy

Volatility suggests AI isn't a bubble...yet

Rising volatility is a clear sign of an asset bubble

Volatility rising along with asset prices is among the clearest signs of an asset bubble. This is due to several factors including asset prices de-coupling from fundamentals and fundamentals themselves becoming more uncertain. Looking at nine asset bubbles since the 1920s, volatility rose in every instance ahead of the market peak.

But tech vol says we're not there yet

US equity momentum near 100yr highs earlier this year and the dominance of just a handful of US tech stocks (the "Magnificent 7" are larger than the 2nd - 5th largest countries by market cap) have some fearing a bubble. However, the lack of a material rise in tech volatility suggests we are not there yet. Return dispersion (how much stocks diverge) also shows few bubble signs as do valuations compared to the late '90s.

The real question is can we avoid a bubble?

The question is can we incorporate AI into the economy without an asset bubble ensuing? It may be hard to avoid given the likely significant but unclear way in which AI will impact the global economy, not dissimilar to the internet in the '90s or railroads in 1840s Britain. The dominance of price momentum, investors remaining believers in buying equity dips, and meme stock popularity only add to the potential for irrationality.

Active managers face historic risk & fragility & it may rise

The proportion of active equity risk emanating from US tech today is at all-time highs, even exceeding the 2000s. If a bubble ensues, this active risk will only rise. At the same time, stock fragility is reaching new heights (recent earnings are driving historic jumps in stock prices), which further adds to the challenges of managing active risk today.

If Y2K is a risk, today looks closer to '95 than '99

While no analog is perfect, the 2000s tech bubble looks most relevant to consider the potential risk of an AI bubble. That bubble took 5yrs to unfold, survived 100bps of Fed hikes, the Asia crisis and a bear-market driven by LTCM's failure, to still generate Nasdaq returns over 80% in each of its final 2 years. Looking at today's returns, volatility & valuations, we are closer to 1995 than 1999.

How to better manage right tail risk

Bubbles are risky to trade. But derivatives can help as they offer participation without needing to time the top & options are still underpricing AI momentum today. Options can also be superior to cash as a risk management tool & reduce the risk of buying dips.

Implications for trading dispersion directly

Dispersion carry trades, while potentially crowded and vulnerable in a correlated macro shock, could have much further to run than many expect if an AI bubble truly inflates.

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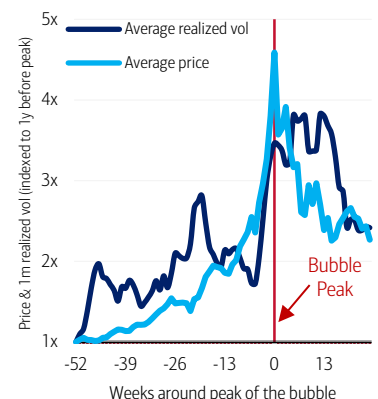
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Exhibit 1: Historically, asset bubbles see a rise in volatility alongside a rise in prices, with vol continuing to rise after the bubble pops

Price and average 1m realized volatility in historical bubbles (vs 1y before peak)



Source: BofA Global Research. Data from Sep-1927 to 09-Feb-22. Historical asset bubbles cover Dow Jones in Sep-29, Gold in Jan-80, Nasdaq in Mar-00, HSCJ in Oct-07, Oil in Jul-08, biotech in Jun-15, Bitcoin in Dec-17 and ARKK in Feb-21.

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AI doesn't look like a bubble yet, but watch risk if it does

FOMO, extreme momentum & US tech dominance feels bubbly. The fear of missing out on upside gains (FOMO) exceeding the fear of downside may seem irrational. But it's happening. Since 2023 US equities have exhibited higher volatility on days equities are up than on days they are down (Exhibit 2). US equity momentum also reached near 100yr records in early 2024 (Exhibit 3). This, plus the sheer dominance of US tech (the "Magnificent 7"¹ stocks have a combined market cap that exceeds the 2nd – 5th largest countries within global equities today; Exhibit 8) has many fearing an asset bubble.²

Rising volatility may be among the best signatures of a bubble. One of the most telling signs of an asset bubble is volatility rising together with asset prices (Exhibit 5). Looking at 9 historical bubbles ranging from US equities in the 1920s to Japanese equities in the 90s, Gold in the 1980s to Bitcoin in 2017, we see a rise in volatility in all of them (Exhibit 4). Volatility rises as asset prices begin to trade purely on momentum and belief in the "greater fool". In equities, volatility rises as prices break free from their tether to fundamentals, and in times of economic disruption (e.g. from the introduction of a new technology), forward visibility of earnings also deteriorates, further supporting volatility.

AI is driving low stock correlation like the 2000s but vol says no bubble yet. The internet bubble which started circa 1995 and peaked in early 2000 may be the best analog to what an AI bubble could look like. The 2000s bubble witnessed both rising stock volatility but also low stock correlation as tech decoupled from "bricks & mortar" companies. Like the 2000s, stock correlation in the US is low today. But volatility has yet to materially rise (Exhibit 7, Exhibit 18), suggesting we may still be far from a bubble.

Active managers beware. The combination of low correlation and rising volatility pushes returns dispersion up (how varied returns are vs a benchmark) which is a direct measure of risk active equity managers face in overweighting/underweighting stocks. If an AI bubble were to develop, the proportion of active risk concentrated in US tech (already at all-time highs, Exhibit 16) could rise even further, making it increasingly challenging to generate alpha outside of these dominant stocks. In fact, fragility (the tendency for assets to see large erratic jumps vs. their "normal" behavior) is today at all-time highs for US large-cap stocks (Exhibit 27), already adding challenges to managing active risk.

Can we escape a bubble? If not, it looks more like '95 than '99. Whether we can escape a bubble is an important question. Jensen Huang, CEO of Nvidia, has called AI the "New Industrial Revolution". The original industrial revolution resulted in several bouts of stock price 'mania' in British railroad companies from the 1830s-1860s. Like in the 2000s, it may be hard for investors to resist pulling forward perceived future growth. Markets today are also prone to momentum chasing. However, it could be years in the making with numerous disruptions along the way. Based on volatility, valuation and returns, it looks closer to 1995 today than 1999. This time may also be different. Today's tech leaders are much larger than in the 2000s (Exhibit 13) and their sheer size may prevent as much decoupling from fundamentals. Nonetheless, the risk markets become more "bubble-like" than they are today should not be ignored in our view.

How to mitigate right tail risk. What does this mean for vol dispersion?

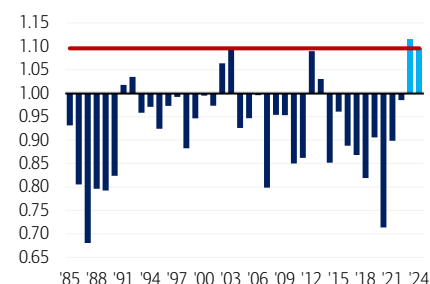
Derivatives can be a key tool for more efficiently trading bubbles. Options are not only underpricing upside momentum in AI-related stocks but also trump cash for managing tail risk (as shown in our report [Don't dash to cash; you have options](#)) and create valuable asymmetry when buying dips (see our 2024 Outlook report, [Flying Blind](#)). Dispersion trades, while potentially crowded and vulnerable in a correlated macro shock, could run much further and for much longer than many expect if an AI bubble truly inflates.

¹ "Magnificent 7" or Mag7 are defined as the group of Apple (AAPL), Amazon (AMZN), Alphabet (GOOGL), Meta (META), Microsoft (MSFT), Nvidia (NVDA) and Tesla (TSLA).

² Near the peak of the infamous Japanese equity bubble in 1989, 7 out of the 10 largest stocks globally were Japanese. See *The New Look Of The Global 1000* on Bloomberg (<https://www.bloomberg.com/news/articles/1999-07-11/the-new-look-of-the-global-1000-intl-edition>).

Exhibit 2: Nasdaq 100 stocks are exhibiting greater "panic" in rallies than sell-offs; historically unusual behavior

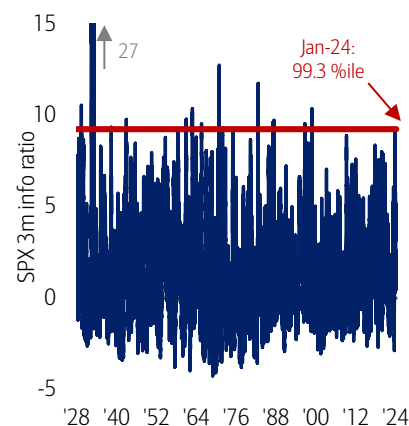
Ratio of NDX realized upside vol vs downside vol



Source: BofA Global Research. Data from 1985 to 30-May-24
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Exhibit 3: In Jan-24, the S&P 500 exhibited near 100yr records in momentum

SPX 500 3m information ratio (return/volatility)

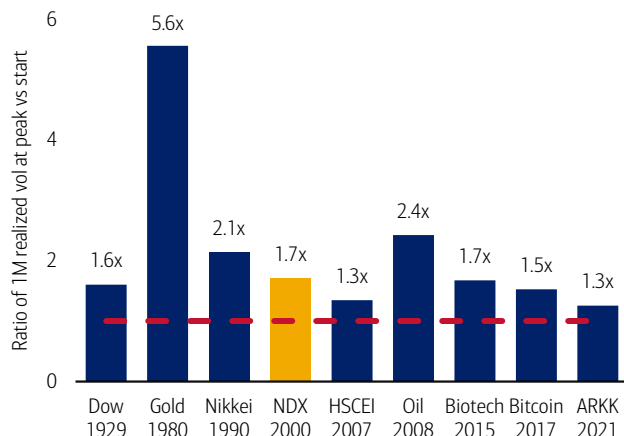


Source: BofA Global Research. Data from Jan-28 to 28-May-24.
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Exhibit 4: Looking at 9 notable asset bubble in the last 100 years, volatility has consistently risen as the bubble developed

Rise in 1m realized vol from start to peak for historical bubbles

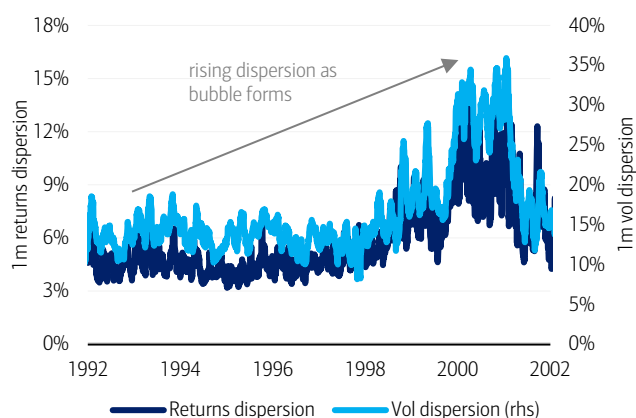


Source: BofA Global Research. Data from Sep-1927 to 09-Feb-22.

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Exhibit 6: The late 1990s saw a rising level of stock dispersion (stock returns diverging from each other) as the tech bubble formed

S&P 500 single stock 1m returns dispersion & 1m vol dispersion

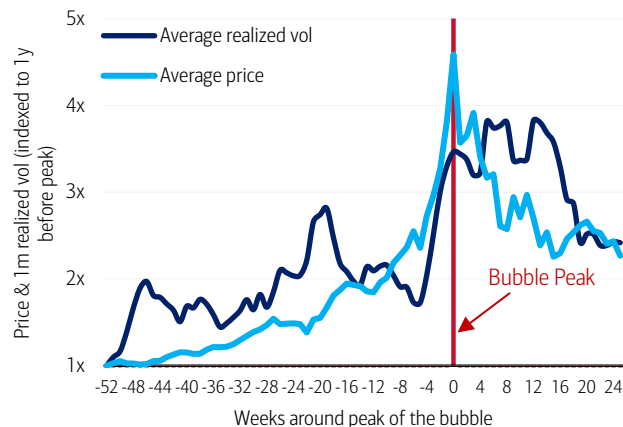


Source: BofA Global Research. Data from 01-Jan-92 to 31-Dec-01.

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Exhibit 5: Historically, asset bubbles have seen a rise in volatility concurrent with a rise in asset prices, with volatility continuing to rise after the bubble pops

Price and average 1m realized vol in historical bubbles (vs 1y before peak)

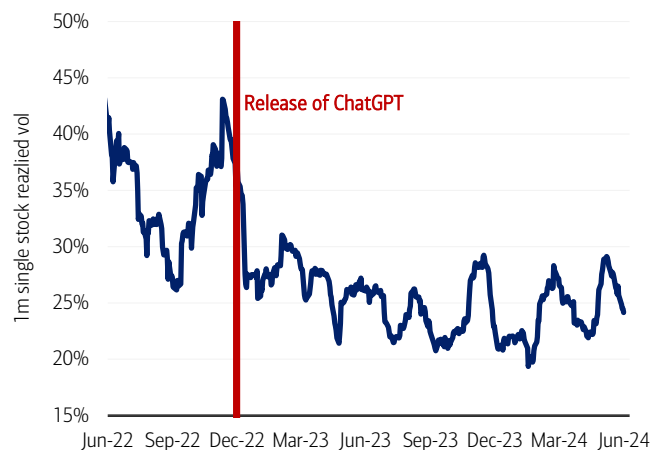


Source: BofA Global Research. Data from Sep-1927 to 09-Feb-22. Historical asset bubbles cover Dow Jones in Sep-29, Gold in Jan-80, Nasdaq in Mar-00, HSCEI in Oct-07, oil in Jul-08, biotech in Jun-15, Bitcoin in Dec-17 and ARKK in Feb-21.

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Exhibit 7: Unlike the tech bubble (and other asset bubbles), single stock vol in the AI era has not started to rise materially since the release of ChatGPT in Nov-22

S&P 500 single stock 1m realized vol



Source: BofA Global Research. Data from 1-Jun-22 to 31-May-24.

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Why should we expect a bubble?

Can markets resist pulling forward future potential?

Technological developments like the internet and everything it brought with it including e-commerce, social-media, cloud computing and now AI (the data to train AI would not exist without the internet), have strong long-term transformational impacts on the global economy. One could argue that the AI revolution is nothing more than an extension of a tech “super cycle” that began in the 90s with the internet with winners of that era continuing to benefit.

In fact, today the US represents over 70% of global equity market cap, up from ~50% 20yrs ago, with nearly all of this gain coming from the increase in value of large-cap US tech companies (Exhibit 8, Exhibit 9). This has also caused the S&P to become more concentrated in fewer large companies than ever before (Exhibit 10).

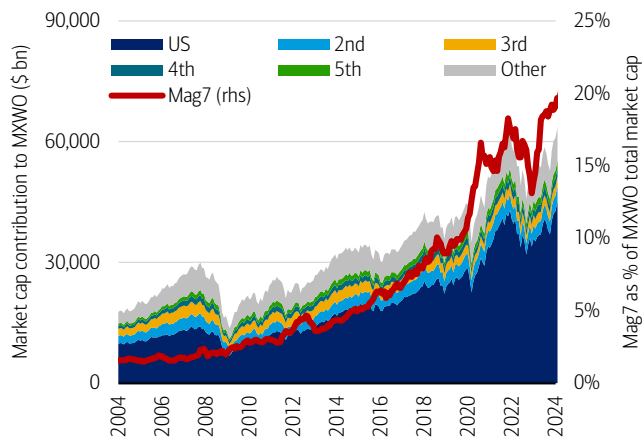
As investors increasingly see a significant future potential in AI, the question is can they avoid the temptation to pull forward the value into asset prices in the form of a bubble? Indeed, even in the 1800s, the British Railway Manias demonstrated this propensity despite indicators showing insufficient railway transport demand. The result was a mega-bubble in the 1840s, which ultimately popped unceremoniously.³

Markets today are prone to momentum chasing, perhaps exacerbating the risks

Markets may be more susceptible to this risk now as momentum has become an increasingly dominant factor today in an alpha-starved world (hedge funds have underperformed the S&P in 5 out of the last 6 years; see our Year Ahead report [Flying Blind](#)). More investors are becoming closet momentum investors (e.g. increasing correlation between discretionary macro hedge funds and trend-following CTAs). Nor is retail shy in chasing irrational price momentum, reminded by the recent re-emergence of meme stock volatility. With conviction on the transformational nature of AI only building, the risk of it turning into a bubble remains real, in our view.

Exhibit 8: Magnificent 7 stocks' market cap is now bigger than the 2nd to 5th largest countries within world equity indices

Geographical breakdown of MSCI World Index constituents

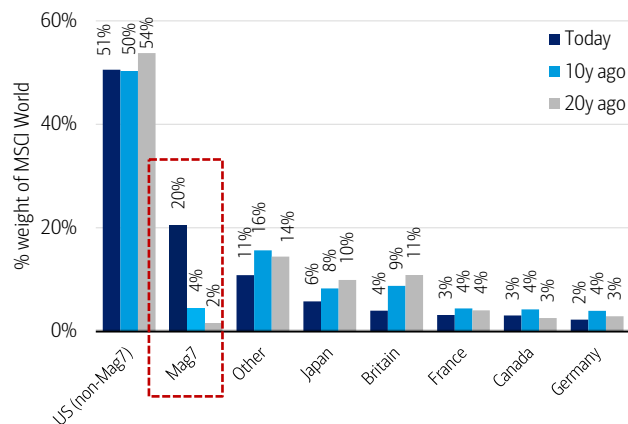


Source: BofA Global Research. Data from 31-May-04 to 22-May-24. “Magnificent 7” or Mag7 are defined as the group of Apple (AAPL), Amazon (AMZN), Alphabet (GOOGL), Meta (META), Microsoft (MSFT), Nvidia (NVDA) and Tesla (TSLA).

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Exhibit 9: The US is now over 70% of global equity market cap, up from 56% 20yrs ago, with nearly all gains driven by large-cap tech

Geographical breakdown of MSCI World Index constituents



Source: BofA Global Research. Data from 31-May-04 to 22-May-24. “Magnificent 7” or Mag7 are defined as the group of Apple (AAPL), Amazon (AMZN), Alphabet (GOOGL), Meta (META), Microsoft (MSFT), Nvidia (NVDA) and Tesla (TSLA).

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³ More details on the British Railway Mania can be found in two of Andrew Odlyzko’s papers: “Collective hallucinations and inefficient markets: The British Railway Mania of the 1840s” (2010) and “The railway mania of the 1860s and financial innovation” (2024)



Exhibit 10: The concentration of market cap within the S&P 500 is far higher today than what has been seen for more than 30 years

Herfindahl-Hirschman Index for the S&P 500



Source: BofA Global Research. Data from 31-Jan-90 to 8-May-24. Herfindahl-Hirschman index (HH index) is a measure of market share concentration and is calculated by summing the squares of the single stock weightings (in percentage points). **Note:** By applying a square (non-linear increasing function) to the constituent weights, the HH index applies more weight to large firms (unlike a concentration ratio), giving a robust picture of concentration.

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The Roaring 90s: Re-examining the internet bubble

The tech bubble emerged over a period of 5 years and withstood two historical macro shocks (the Asian crisis in 1997 and Russia's default leading to Long Term Capital Management's demise and bailout in 1998) along with a 100bps of Fed hikes towards the end (Exhibit 11). The fallout from LTCM led to a near bear-market in the S&P 500 (just shy of 20%) and a bear market in the Nasdaq 100 (-23% drawdown). Nonetheless, over this period, the Nasdaq experienced extraordinarily fast-paced returns (85% in 1998 and over 100% in 1999; see Exhibit 12).

Why this time may be different

To the extent that the tech bubble is a useful comparison to today's AI-driven bull market, 2023 most resembled 1995 or 1996 in terms of return and drawdowns, though with lower levels of vol.

Size may matter

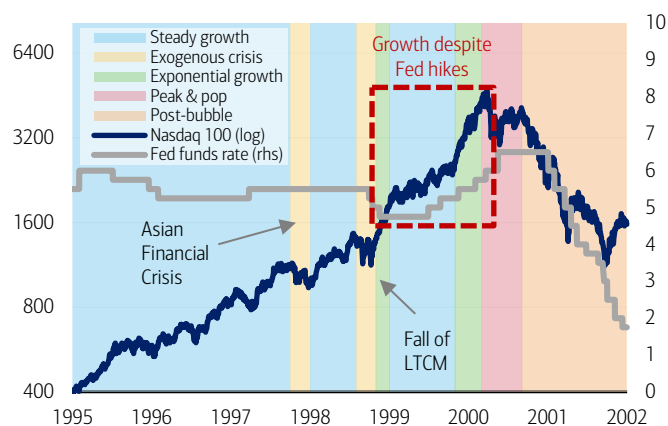
However, one major difference to today is tech companies' sheer size, which might act as a hurdle in achieving a comparable valuation expansion. Indeed, today's tech megacaps are close to 3 times larger than those of the tech bubble when compared to the size of the US economy (Exhibit 13). Note this did not stop stocks like Cisco in the 2000s (sometimes compared with Nvidia today) from rising from a reasonable PE of ~35 in 1995 to an extreme PE of nearly 400 at the 2000s peak. Moreover, if AI enthusiasm expands to a larger set of smaller companies in future phases of the technological evolution, the process of bubble formation would be less inhibited by the size of the megacaps.

Narrow leadership reminiscent of a bubble

As has been pointed out by our Chief Investment Strategist, Michael Hartnett (see [31-May-24 The Flow Show](#)), US equity market breadth is currently low, in other words there are only a small number of stocks driving recent gains. As noted in Exhibit 14, 2024's poor breadth (during a rising market) has only been exceeded in 2007 and 1998, both prior to asset bubbles popping. However, one has to also consider the backdrop today with extreme market-cap concentration owing to the secular growth in US large-cap tech (Exhibit 9, Exhibit 13). If the initial AI winners turn out to be those most successful in the last 20 years, the risk is that this narrowness continues.

Exhibit 11: Between 1995 & early 2000, the Nasdaq grew almost 10x while withstanding the shocks of the Asian Financial Crisis and the fall of LTCM, as well as 100bps of Fed rate hikes

Nasdaq 100 prices (log scale) over the tech bubble era

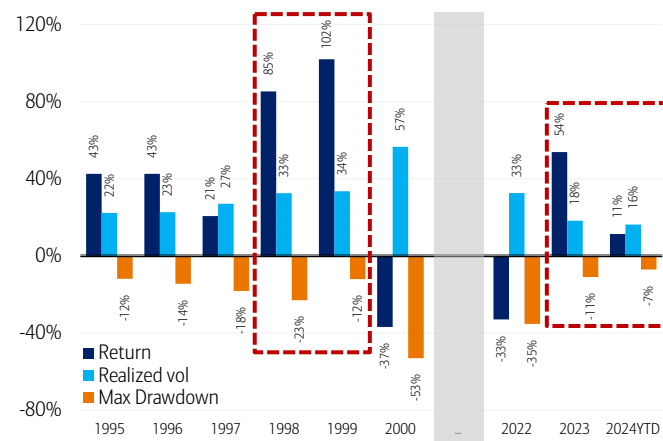


Source: BofA Global Research, Bloomberg. Data from 1-Jan-95 to 31-Dec-01.

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Exhibit 12: Recent price action has been muted compared to what was seen during the tech bubble, particularly relative to the explosive growth witnessed in 1998-99

Nasdaq 100 returns, realized vol & max drawdown for the tech bubble (1995-2000) & the AI wave (2022-)



Source: BofA Global Research. Data from 1-Jan-95 to 21-May-24. Metrics are calculated for each calendar year.

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Have investors learned a lesson?

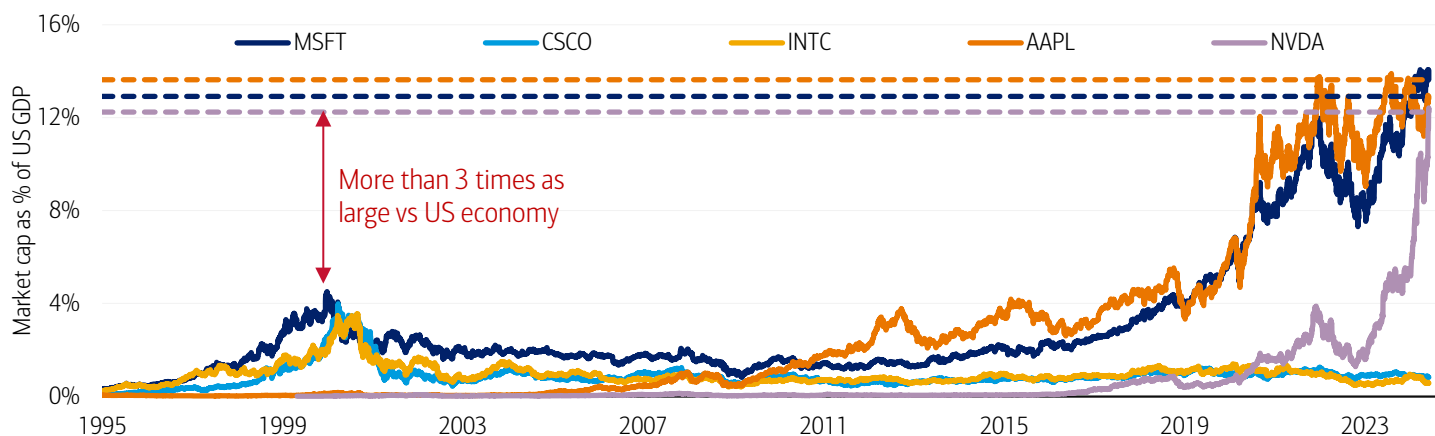
Lastly, it's also possible investors have learned from recent mistakes with two of the largest asset bubbles in 100 years occurring in the last 30 (2000s tech bubble and the 2007-8 housing bubble). However, as noted above, we have also seen a higher propensity to chase momentum recently, not to mention meme stock popularity, raising the question of whether investors are truly less susceptible to this risk today.

What about the Fed?

Remarkably, even the Fed hiking rates by 100bps in the late stages of the 2000s tech bubble (Jun-99 to Feb-00) did not stop the bubble from inflating (Exhibit 11). While economic uncertainty today remains high and markets have been consistently wrong-footed in their forecasts for rate cuts, most still believe the next move will be lower in rates, which would perhaps only support the market further, though some argue this could actually be a trigger for markets to sell-off.

Exhibit 13: Compared to the tech bubble, today's tech megacaps are almost 3 times larger with respect to the entire US economy, which may act as a hurdle in the pace at which an AI bubble forms

Market cap as a percentage of US GDP for the Top 3 tech companies in the tech bubble and today

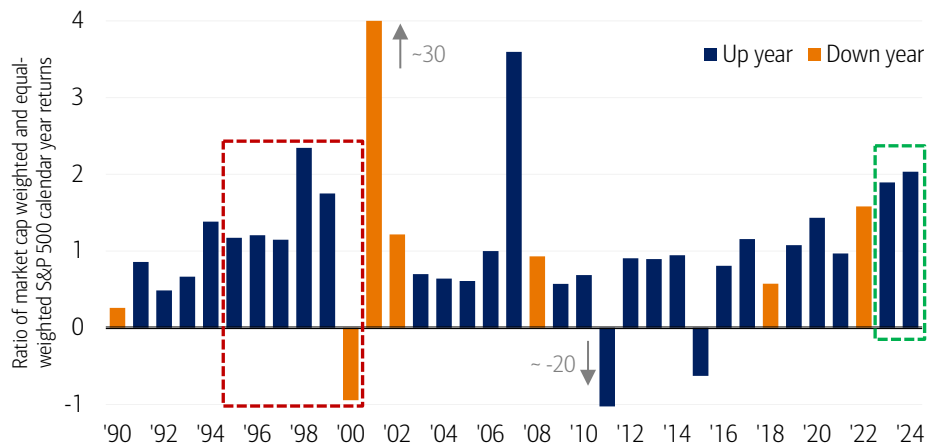


Source: BofA Global Research. Data from 1-Jan-95 to 28-May-24.

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Exhibit 14: The breadth in the S&P's rally this year has been historically low (led by a small number of stocks) and was only exceeded in 2007 & 1998, each near the end of historical asset bubbles

Ratio of market cap weighed and equal-weighted S&P 500 calendar year returns



Source: BofA Global Research. Data from 1-Jan-90 to 31-May-24. "Up year" and "Down year" defined by when market-cap weighted S&P 500 has positive or negative calendar year returns.

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Returns dispersion and its implications in asset bubbles

Perhaps one of the most striking signatures of the tech bubble is the historically high levels of returns dispersion exhibited by equities (Exhibit 15). In fact, dispersion levels around the tech bubble peak were higher than even the Global Financial Crisis of 2008. This contrasts with the picture today, where dispersion levels are quite far from tech bubble levels. Given such a striking picture and its implications for active investing, we find dispersion to be an interesting and useful lens to view bubble formation through.

What is returns dispersion?

Returns dispersion can be seen as a measure of how much index components' returns vary vs the aggregate index return. In this analysis, we measure returns dispersion within the S&P 500 as the weighted average of the absolute differences of single stock returns and index returns (with weights as per the stocks' weightings within the index). More simply, returns dispersion measures the average deviation of single stocks from the index.

Why does it matter? It's a direct measure of active manager risk

Returns dispersion can be seen as a direct measure of active risk for portfolio managers. Periods of low dispersion, where stock returns are similar in direction and magnitude to each other (and hence the index), have lower active risk, making stock picking less crucial but also increasing the appeal of passive investing. Conversely, periods of high dispersion, where stock returns vary in direction and magnitude to each other, have higher active risk and create greater risk (and opportunity) for stock pickers.

With the tech bubble exhibiting historically high levels of dispersion, the formation of an AI bubble could greatly exacerbate active risk in equity markets. With active risk emanating from tech already historically high today mainly owing to high tech concentration, (Exhibit 16), an AI bubble in which stock volatility and market caps rise further would only make such risk more extreme.

Why was returns dispersion so high in the tech bubble? Why not now?

Since returns dispersion looks at the magnitude and direction of single stock returns, single stock volatility and correlation levels are the primary drivers, with high volatility and low correlation contributing to high dispersion. The tech bubble, particularly in the latter stages, was witness to both of these trends concurrently (Exhibit 18, Exhibit 19), driving returns dispersion to historically high levels. As one would expect, much of this dispersion was driven by an increase in dispersion in tech (Exhibit 16).

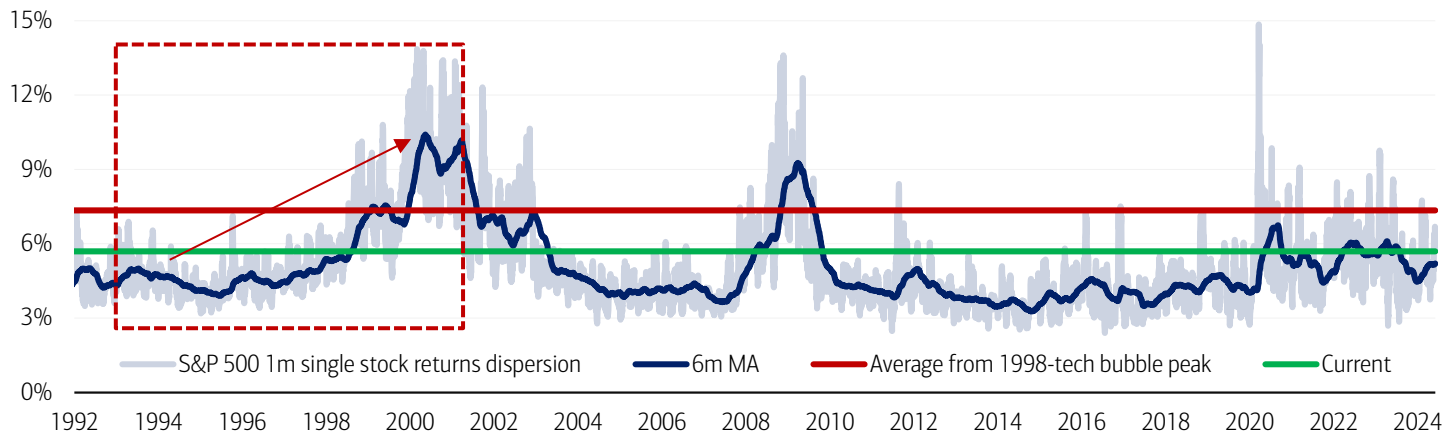
By contrast, today's AI rally, while seeing a sharp drop in stock correlation (Exhibit 19), has not seen the same kind of rise in single stock vol as the tech bubble (Exhibit 18). While recent months have seen somewhat of an uptick in vol, and the post-Covid vol levels have been higher than those in the 2011-2019 period (most likely driven by higher macro risk), current dispersion levels are still a far cry from those seen during 1998 to 2000, even within the tech sector mainly owing to much lower stock vol.

Notwithstanding this, in our view, the potential for an AI bubble to drive a rise in stock vol (and consequently returns dispersion, or in other words active equity risk) remains real, particularly given how much the dispersion risk on the index level is being driven by AI-exposed megacap tech.



Exhibit 15: The tech bubble saw returns dispersion (a direct measure of active equity risk) rise to historically high levels, more so than even in the GFC in 2008; while current levels are not historically low, they are a far cry from those seen in the tech bubble

S&P 500 1m single stock returns dispersion

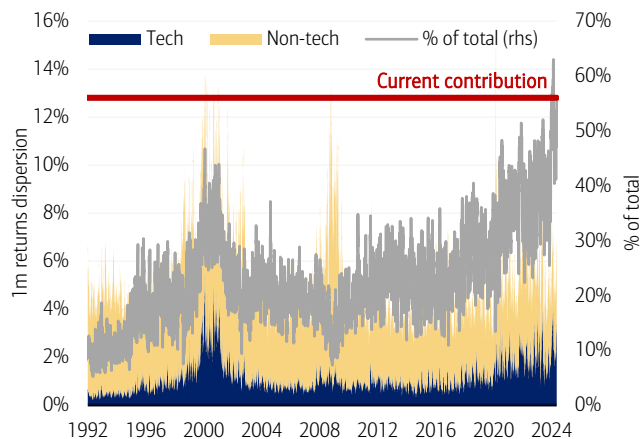


Source: BofA Global Research. Data from 1-Jan-92 to 31-May-24. Returns dispersion is calculated as the weighted average of the absolute differences of single stock returns and index returns (with weights as per the stocks' weightings within the index).

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Exhibit 16: Today, tech's contribution to returns dispersion (active equity risk) in the S&P is close to all-time highs and higher than its (extraordinary) contribution during the tech bubble...

Tech contribution to S&P 500 single stock 1m returns dispersion (market cap weighted)

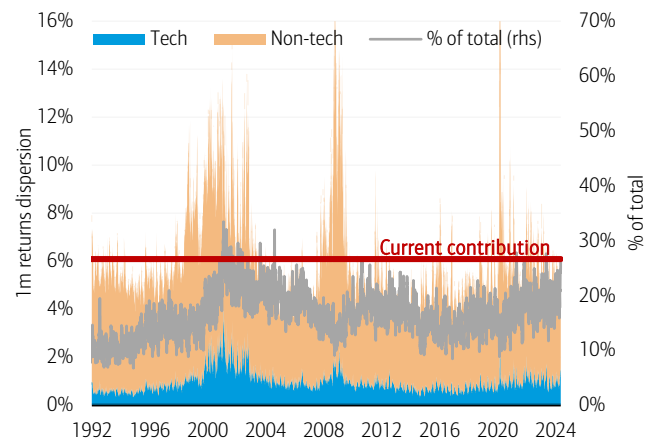


Source: BofA Global Research. Data from 1-Jan-92 to 31-May-24. "Tech" is defined by the Information Technology class of the Global Industry Classification Standard along with companies in other classifications that are primarily internet/tech driven (e.g. Tesla, Amazon, Uber etc.).

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Exhibit 17: ...but much of this is driven by increasing market cap concentration, since contribution on an equally weighted basis has only risen moderately

Tech contribution to S&P 500 single stock 1m returns dispersion (equally weighted)

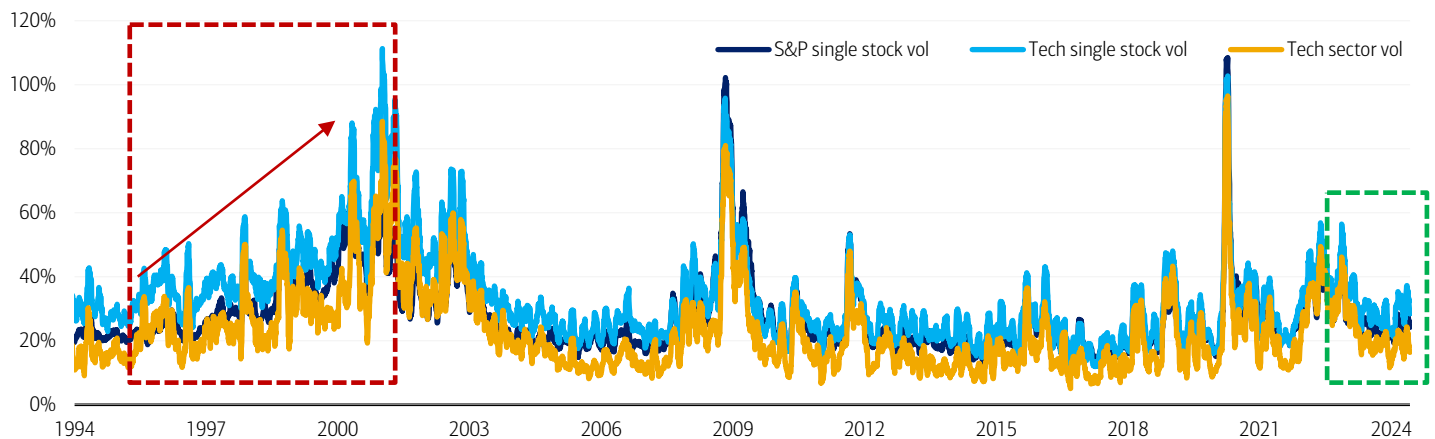


Source: BofA Global Research. Data from 1-Jan-92 to 31-May-24. "Tech" is defined by the Information Technology class of the Global Industry Classification Standard along with companies in other classifications that are primarily internet/tech driven (e.g. Tesla, Amazon, Uber etc.).

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Exhibit 18: While vol, both in tech & the wider S&P index, rose as the tech bubble formed, it has yet to trend materially higher since the release of ChatGPT in Nov-22

Single stock 1m realized vol for S&P 500 tech stocks & wider index; realized vol for tech returns

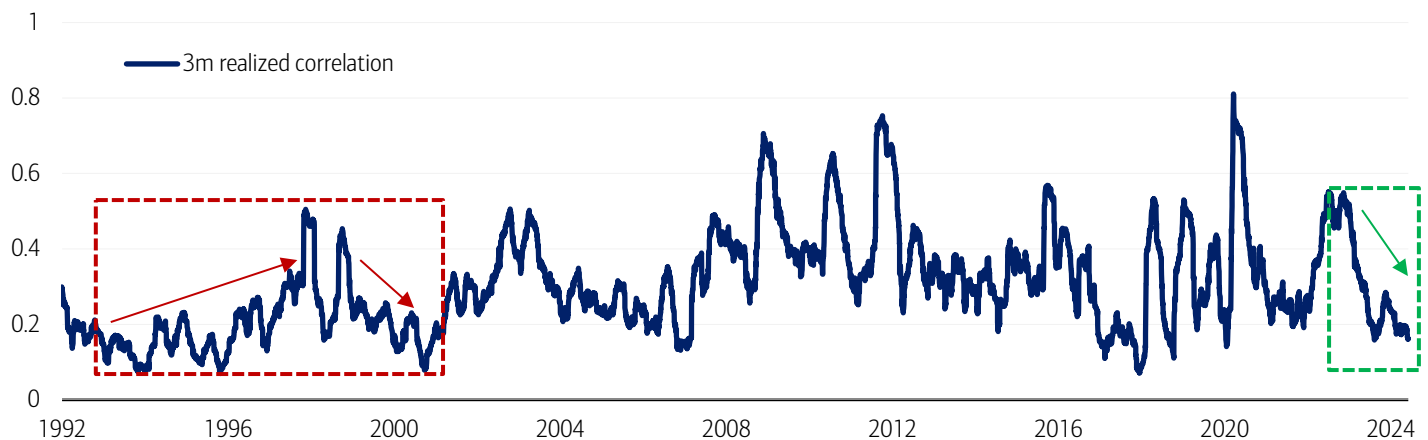


Source: BofA Global Research. Data from 1-Jan-95 to 31-May-24. "Tech" is defined by the Information Technology GICS classification along with primarily tech-driven stocks in other classes (e.g. TSLA, AMZN, etc.). Single stock vol is calculated as the weighted average of the volatilities of S&P 500 constituents (based on prevailing index weights).

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Exhibit 19: Today's single stock correlation has been on the decline, similar to the latter stages of the tech bubble. However, initial stages of the bubble saw rising single stock correlation within the index

S&P 500 3m realized single stock correlation



Source: BofA Global Research. Realized correlation data from 1-Jan-92 to 31-May-24. Single stock correlation is derived from single stock vol and index vol levels.

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Volatility in bubbles

Why should tech volatility rise as AI develops?

Momentum breaks fundamental tether

As asset bubbles form, a key reason volatility rises is that stocks start trading purely on momentum, decoupling from their fundamental tether (where fundamentals exist⁴). Exhibit 20 shows how this tether was broken in the Nasdaq in 1998-2000 (vs the S&P), in contrast to how closely linked fundamentals & tech stock prices appear today.

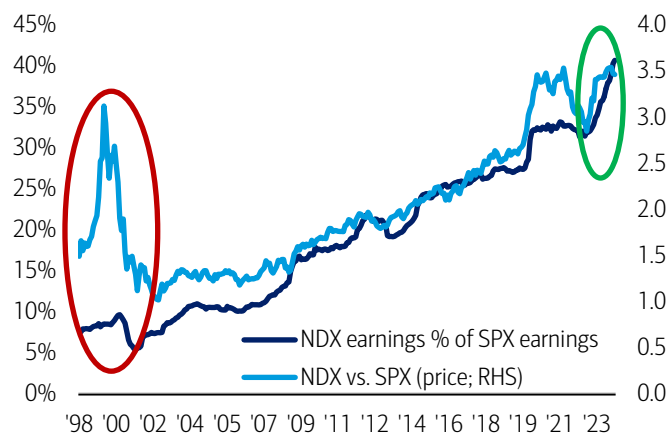
Fundamentals become more uncertain

Economic disruption by nature creates uncertainty for corporate business models and hence earnings. AI is already creating uncertainty for what the business models of even dominant tech leaders will look like in the future. Will Large Language Models (LLMs) change the nature of internet search which Google currently dominates, or the mode of ad monetization? The consequent difficulty in forecasting future earnings creates greater valuation uncertainty, contributing to higher levels of volatility.

While such uncertainty has not trickled into pushing stock volatility to tech bubble levels yet, it has arguably contributed to an increase in stock fragility that we have most recently witnessed around earnings releases (see Exhibit 26 to Exhibit 28). There are likely other reasons stocks have become more fragile as well, including trade crowding, reflexive behavior (price action driving fundamental beliefs, which can become a self-fulfilling feedback loop), and fickle trading liquidity. These are all factors we believe have led to financial markets more broadly becoming fragile in recent years.⁵

Exhibit 20: During the tech bubble stock prices clearly decoupled from fundamentals, a phenomenon we have yet to see recently

Nasdaq 100 vs S&P 500 – earnings & performance

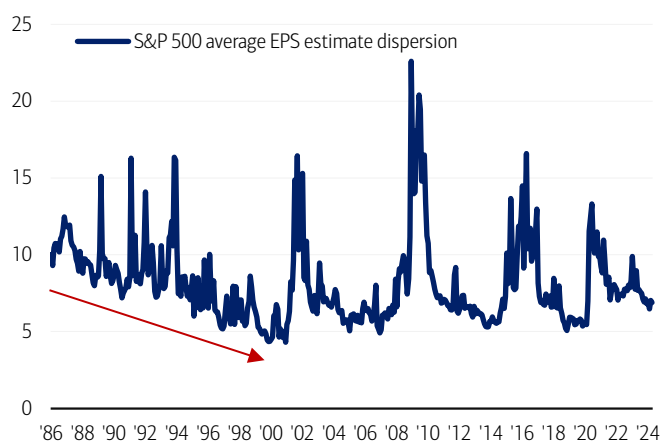


Source: BofA US Equity & Quant Strategy, FactSet. Data from Jan-98 to Apr-24.

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Exhibit 21: Oddly, during the tech bubble the dispersion of analysts' earnings estimates declined, despite fundamental uncertainty rising, a trend we have also seen recently⁶

S&P 500 average EPS estimate dispersion



Source: BofA US Equity & Quant Strategy. Data from Jan-86 to Apr-24. Estimate dispersion is the market cap weighted avg of standard deviation of estimates (normalized by mean estimate).

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More 'surprise' as analysts stick together as uncertainty rises?

Somewhat strangely, during the internet bubble, as fundamental uncertainty rose and stocks de-coupled from fundamentals, the dispersion in analysts' earnings estimates declined (Exhibit 21). This seems the opposite of what one would expect, however, perhaps it's an artifact of high uncertainty causing analysts to avoid straying too far from consensus. The juxtaposition of what appears to be higher certainty in earnings

⁴ Arguably assets such as Gold and Bitcoin have much more difficult to pinpoint fundamentals which may allow for momentum to dominate price action more easily, as the 'fundamental tether' is less defined than for assets like stocks.

⁵ We first outlined this phenomenon in our 2016 Year Ahead Outlook [Fragility is the new volatility](#)

and the reality of rising earnings uncertainty may result in larger earnings surprises. Interestingly, the recent trend has also been for earnings dispersion to decline.⁶

Why is vol, specifically tech vol, not higher today?

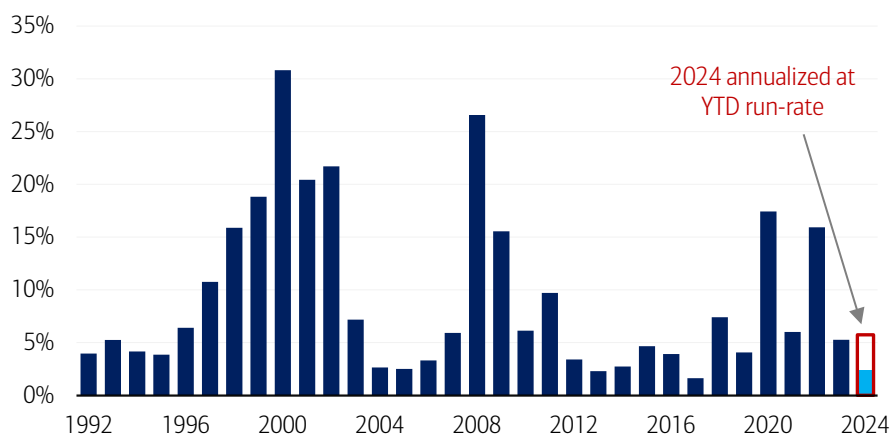
While we have started to see an early rise in single stock vol (and fragility as noted), the trend since ChatGPT was launched has been that of lower, or at least not rising, vol (even in tech; see Exhibit 18). Some of the decline is likely due to vol falling post the stress from the 34% decline in Nasdaq during 2022. However, vol would not be categorized as particularly low either, and the question of why vol is not higher and closer to tech bubble levels is one worth exploring.

Stocks are moving, but just not enough

Mechanically, higher vol levels would require stocks to have more extreme moves on a frequent basis. As Exhibit 22 demonstrates, using a returns magnitude threshold like 3%,⁷ the largest stocks in the S&P are just not moving as much today as they used to in the late 90s.

Exhibit 22: While the frequency of large daily returns among large S&P stocks has increased compared to pre-Covid, they are not close to as extreme as witnessed during the tech bubble

Frequency of single stock daily returns greater than 3% in the largest 50 stocks in the S&P 500



Source: BofA Global Research. Data from 01-Jan-92 to 31-May-24. Frequency is calculated as the number of instances when any of the largest 50 stocks in the S&P realized a 1d return magnitude of greater than 3% divided by 12,600 (252 days x 50 stocks moving >3%).

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Tech appears immune to macro risk

One of the key reasons for stocks moving less than one might expect lies in tech's seeming immunity to macro risk. Despite forward visibility on monetary policy remaining low and economic volatility remaining historically high, tech seems to have exhibited an asymmetric reaction function to macro risks, cheering for the positives and remaining muted for the negatives (Exhibit 23). This could be a sign of how underweight investors have been and how afraid they are to sell stocks with the kind of momentum tech has shown.

Quite contrastingly, small caps have exhibited a far more symmetric reaction function (Exhibit 24). Such an asymmetric reaction function in many of the biggest stocks can contribute to keeping aggregate stock vol levels contained. Note this is a distinct difference from before ChatGPT's launch when in most of 2022, tech (arguably a long duration asset) was strongly negatively correlated to rates.

⁶ Note that historical levels may not be comparable to present day ones due to the introduction of Regulation Fair Disclosure in October 2000.

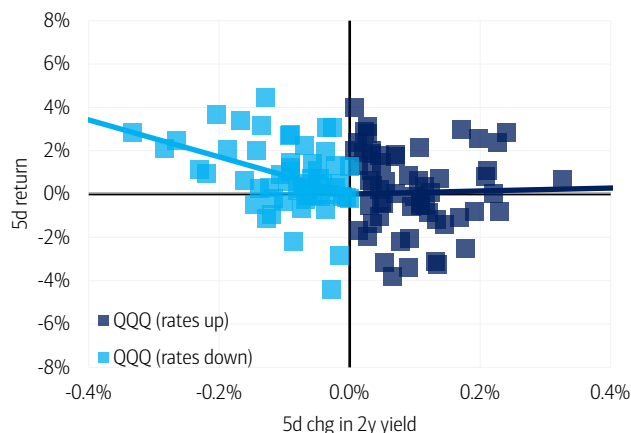
⁷ We choose a 3% daily returns threshold since it approximately corresponds to 50% annualized volatility, which is the level of single stock volatility seen through numerous periods in the tech bubble



Investors have also exhibited a strong propensity to buy equity dips, arguably a learned response to years of central banks teaching them it's the right trade (Exhibit 25). This may also help reduce downside versus upside volatility which as noted above has been a notable feature of markets in recent years.

Exhibit 23: The tech heavy Nasdaq has been asymmetric in reacting to rates, rallying hard when rates fall but staying more muted when rates rise...

5d QQQ returns vs 5d changes in 2y yields since Dec '23 Powell pivot

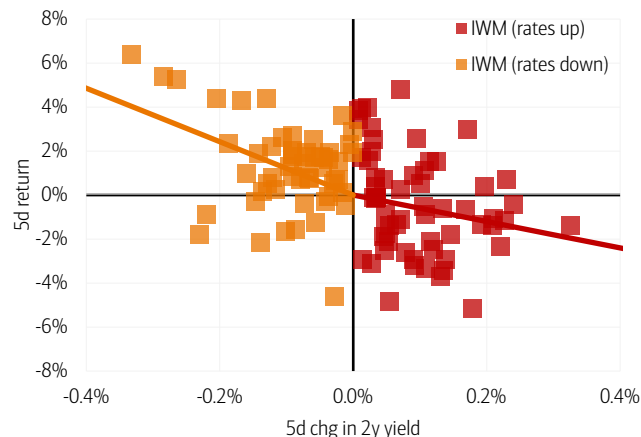


Source: BofA Global Research. Data from 13-Dec-23 to 21-May-24.

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Exhibit 24: ...while the small cap Russell 2000 has shown a relatively symmetric rates up, equities down reaction function

5d IWM returns vs 5d changes in 2y yields since Dec '23 Powell pivot

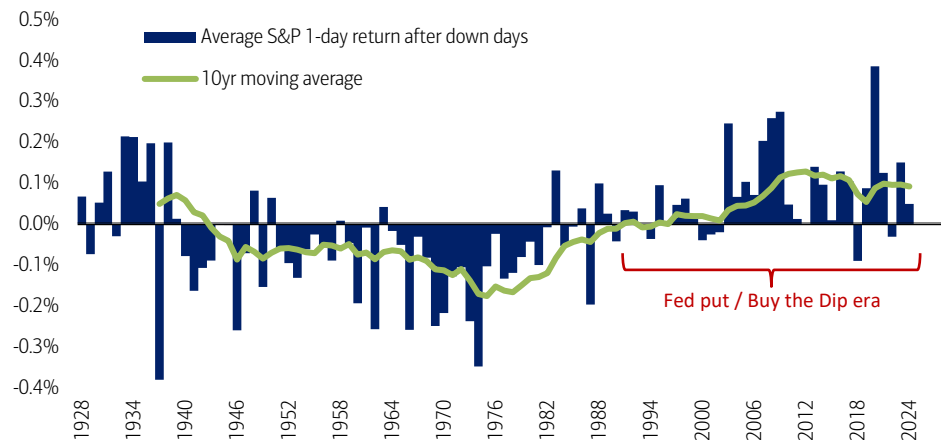


Source: BofA Global Research. Data from 13-Dec-23 to 21-May-24.

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Exhibit 25: While buy-the-dip trends have been generally strong since the introduction of the Fed put in the late 80s, they have been particularly so over the past 2 decades

S&P 500 1-day returns after down days (calendar-year average)



Source: BofA Global Research. Data from Jan-1928 to 31-May-24.

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Today's large cap tech have stronger profiles

More fundamentally, today's tech companies are more financially stable than those in the tech bubble, and stocks are still tethered to their fundamentals. Tech valuations remain substantially lower than those in the late 90s, particularly in megacaps where market cap concentration is increasing, with earnings growth remaining strong (Exhibit 20).

What about vol selling flows?

Lastly, option selling strategies like call overwriting or autocallables could in theory be suppressing single stock volatility today (particularly compared to the 1990s tech bubble when such strategies were likely less popular). While more work remains to be done to

quantify the impact of this vol supply at the single stock level, our analysis at the index level (see our weekly [Systematic Flows Monitor](#) report) shows that such flows tend to be the “icing on the cake” rather than the cake itself in terms of driving S&P 500 realized volatility. Speculative call *buying* has also become a popular use case for single stock options in recent years and could be offsetting the vol dampening impact of call overwriting flows to some degree.

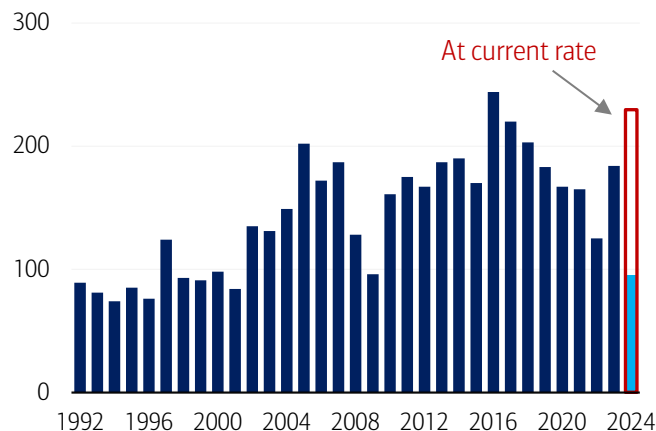
Vol may not be high, but fragility is

While volatility has remained muted, US tech stocks are exhibiting unusual levels of fragility. Fragility is the idea of an asset exhibiting extreme returns relative to what is projected based on the asset’s recent price volatility. It’s akin to witnessing extreme “value-at-risk”, or fat tails relative to what recent volatility would project the return distribution should look like.

Tech stocks this year are on pace to record one of the highest frequency of fragility events over the past 30 years (Exhibit 26), and the 50 largest stocks in the S&P have so far exhibited the most extreme magnitude of these events on average over this period (Exhibit 27 & Exhibit 28). Indeed, tech megacaps like NVDA, GOOGL and META – despite having in some cases trillion dollar plus market caps – have seen double digit returns in the past few months, resulting in multi-hundred-billion-dollar gains & losses (Exhibit 28).

Exhibit 26: We have seen an increasing frequency of fragility events in the past two years, particularly in tech

No. of fragility events (sigma move magnitude > 3) for S&P 500 tech stocks

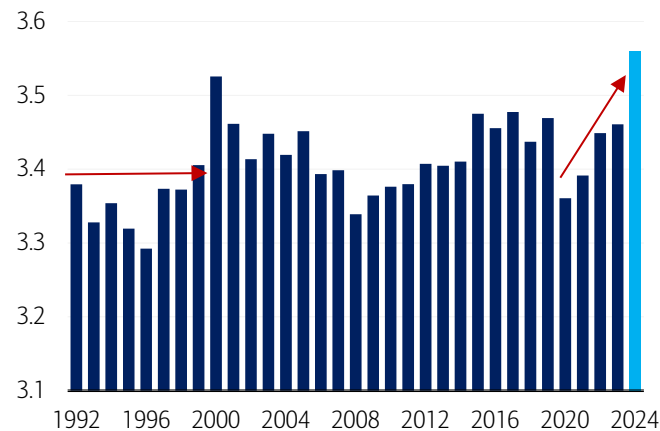


Source: BofA Global Research. Data from 1-Jan-92 to 31-May-24. “Sigma move” is measured by dividing a stock’s 1d return by its 21d trailing volatility. It is a measure of how large a shock is vs its risk projected by recent volatility and is one measure of price “fragility”. “Tech” is defined by the Information Technology GLCS class along with stocks in other classes that are primarily tech-driven (e.g. Tesla, Amazon, etc.). Components of index kept fixed from the start of every calendar year.

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Exhibit 27: Fragility events have also become increasingly more extreme in their magnitude within the largest stocks in the S&P

Average magnitude of fragility events (when sigma move magnitude is > 3) for the largest 50 stocks in the S&P 500



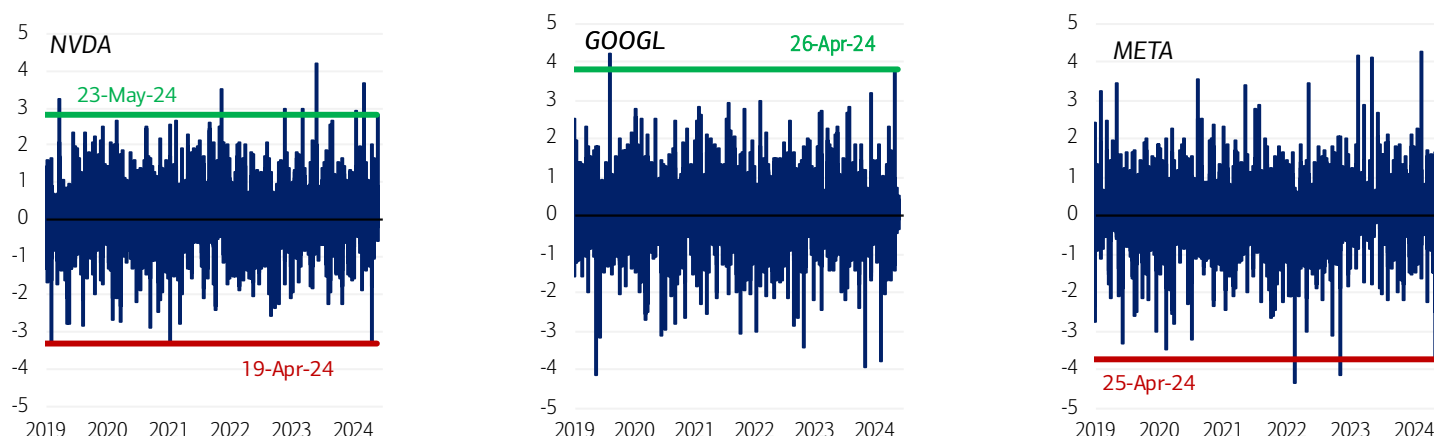
Source: BofA Global Research. Data from 1-Jan-92 to 31-May-24. “Sigma move” is measured by dividing a stock’s 1d return by its 21d trailing volatility. It is a measure of how large a shock is vs its risk projected by recent volatility and is one measure of price “fragility”. Components of index kept fixed from the start of every calendar year.

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Exhibit 28: Bouts of fragility in megacap tech (especially around earnings) have become more frequent of late, highlighting the uncertainty in estimating the value of technological disruption. Other potential structural factors causing markets to be more fragile today may also be at-play

1d returns / 21d trailing realized vol for NVDA, GOOGL and META



Source: BofA Global Research. Data from 01-Jan-19 to 22-May-24.

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Is fragility the new volatility?

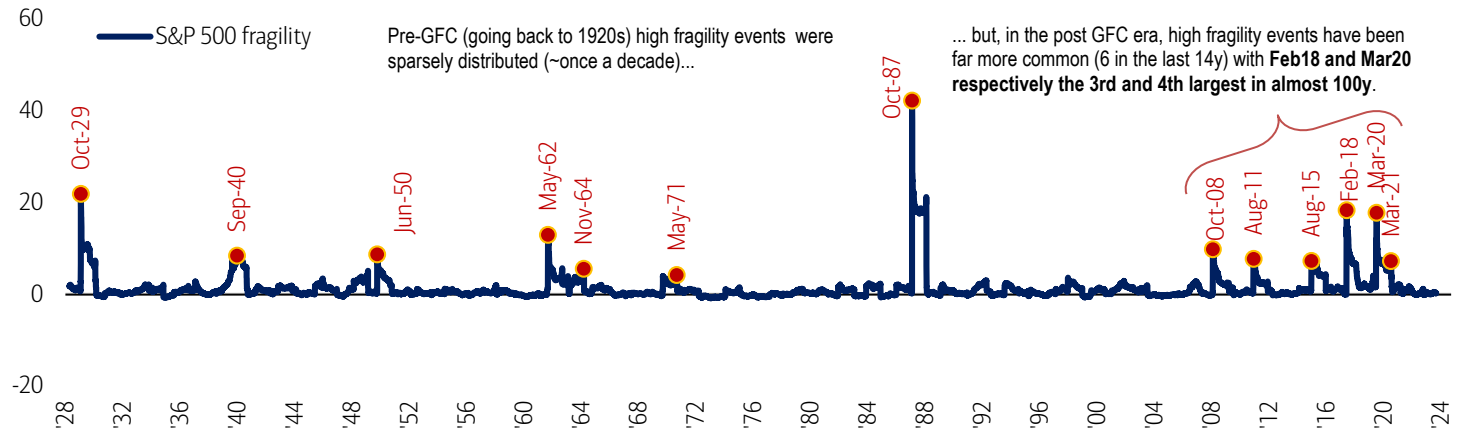
Markets overall have become increasingly fragile not just in equities but across asset classes. The S&P 500 has witnessed a 6-fold increase in the frequency of notable fragility events since the GFC in 2008 compared to the 80 years prior (Exhibit 29). We have argued this owes to structural factors including an over dependence on central-bank policy, volatility suppression, starvation of the markets for alpha, and increasingly fickle trading liquidity emanating from a greater role of high-frequency market-makers.

In our 2015 paper [Fragility is the new volatility](#), we first argued that fragility had become a more important risk factor for active managers than volatility. Hence a reasonable question is whether today's elevated fragility in US tech stocks is simply a more modern indication of a developed asset bubble?

While position crowding seems partly to blame for today's heightened fragility, not to mention secular shifts in investor susceptibility to reflexivity and fickle trading liquidity, the fact stocks remain more grounded to fundamentals today compared to the 2000's suggests there is more room to run in terms of bubble formation. In fact, if volatility were to rise, fragility may somewhat subside, as fragility is a relative risk measure.

Nonetheless it's also fair that one should consider the secular changes to markets in the last 30 years when considering the usefulness of the 2000s period as a potential analog to what a bubble could look like today.

Exhibit 29: Since the GFC in 2008, the S&P 500 has witnessed a 6-fold increase in the frequency of notable fragility events compared to the 80 years prior
S&P 500 fragility since 1928



Source: BofA Global Research, Bloomberg. Data from 31-Dec-1927 to 31-May-24. Fragility = propensity of the S&P 500 to exhibit rate but relatively sharp shocks, measure via rolling 1y kurtosis of weekly returns.

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(De)correlation in bubbles

Correlation is low now, but was it back then?

One of the primary drivers of the low levels of index vol (e.g. S&P 500 and hence VIX) today has been low stock correlation, which has reached levels rarely seen over the past 20 years. However, while these levels are low even over a longer history, stock correlation during the tech bubble appears as low (if not lower at times), particularly in the lead up to the peak (Exhibit 19).

However, while today's downtrend in correlation has been persistent, the initial stages of the tech bubble saw a marked rise in correlation until 1997. One way to rationalize this is to view the technological breakthrough of the internet as one that 'lifted all boats', creating widespread optimism and a risk-on sentiment.

It is only post-1998 that we see signs of decorrelation within stocks. While the correlated risk-off selloffs triggered by the Asian Financial Crisis and fall of LTCM complicate this trend, the downtrend in correlation between 1998 and the peak of the bubble in 2000 appears more akin to today's decorrelation, in our view.

Why so decorrelated?

While on face value the decorrelation today looks similar to that seen towards the end of the tech bubble, digging deeper into the drivers of this metric reveal somewhat different trends under the surface.

Inter-sector correlation is dropping steadily

The sector decorrelation we see today is similar to what we saw during the tech bubble (Exhibit 30 & Exhibit 31). A straightforward hypothesis on the driver of this trend is a "tech vs non-tech" dynamic: stocks directly exposed to the technological breakthrough (tech stocks) rally at the expense of those who aren't (non-tech "brick-and-mortar" stocks).

But today's intra-tech stock correlation is also dropping

Single stocks *within* the tech sector are becoming less correlated with each other today, a trend we didn't particularly see during the tech bubble (Exhibit 32 & Exhibit 33). Such a difference may suggest that today's correlation structure is different than that of the tech bubble, and a "tech vs non-tech" dynamic might not be the only one at play.

Today's high idiosyncratic risk may be a bigger driver than sectoral divergences

Indeed, the secular single stock decorrelation (even within the AI raptured tech) suggests that idiosyncratic risk levels have risen over the past 2 years. The impact of idiosyncratic risk is evident even in the tails: measuring correlation of the largest 50 stocks in the S&P on days of material fragility events, we see a sharp drop over the last 2 years (though we concede that the statistical precision of this measure is low given the infrequency of such events; see Exhibit 34).

In other words, the rise in single stock fragility, which seems to be happening independently and is by some measures reaching 30+yr highs, may also be contributing to the unusually low intra-stock correlation today.

Exhibit 30: Similar to the decorrelation between sectors we have seen since the start of 2023...

S&P 500 3m realized sector correlation

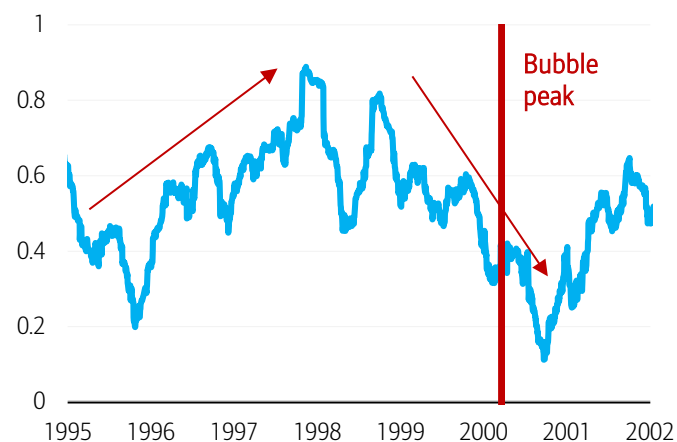


Source: BofA Global Research. Data from 1-Jan-22 to 31-May-24.

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Exhibit 31: ...we saw sector decorrelation towards the end of the tech bubble after the fall of LTCM as well

S&P 500 3m realized sector correlation



Source: BofA Global Research. Data from 1-Jan-95 to 31-Dec-01.

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Exhibit 32: Unlike the single stock decorrelation seen within the tech sector today...

3m realized single stock correlation within tech sector of S&P 500

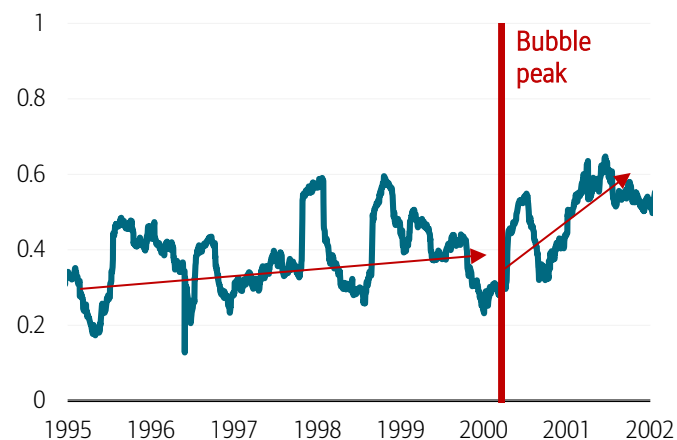


Source: BofA Global Research. Data from 1-Jan-22 to 31-May-24.

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Exhibit 33: ...tech in the late 90s didn't particularly see any similar trends in lowering correlation

3m realized single stock correlation within tech sector of S&P 500

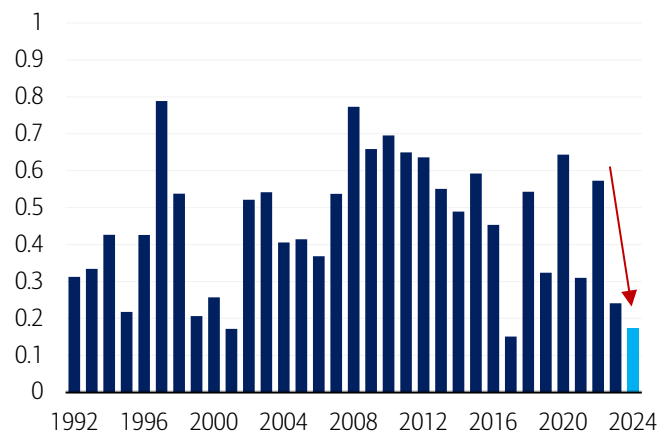


Source: BofA Global Research. Data from 1-Jan-95 to 31-Dec-01.

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Exhibit 34: Idiosyncratic risk in single stocks seems to have risen even in the tails as fragility events get increasingly decorrelated in the largest stocks in the S&P

Calendar-year average pairwise sigma move correlation between the largest 50 S&P stocks on days with fragility events*



Source: BofA Global Research. Data from 1-Jan-92 to 7-May-24. *Days with fragility events are defined as the days in a calendar year when at least 1 stock in the S&P Top 50 experiences a sigma move magnitude > 3 (calculated vs trailing 21 d realized vol). Components of index kept fixed from the start of every calendar year.

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If a bubble is a risk, what can you do about it?

If a 2000s-type bubble is a potential risk for markets in the next several years, what can you do about it? By nature, an asset bubble implies prices escaping from what is fundamentally justified, potentially for such extended periods of time that investors can't avoid participating. As Chuck Prince, famously said in July 2007, "As long as the music is still playing, you've got to get up and dance."

It's also true that bubbles persist not because investors can't identify they are in one, but due to the inability to time the end. This is why options and volatility-linked strategies can be valuable in trading bubble-like markets, as they afford one to participate in the upside without having to time the top. Of course that comes at a cost, but interestingly we find options often underprice asset momentum, almost by construction.

Optionality can help to safely navigate risk of AI euphoria

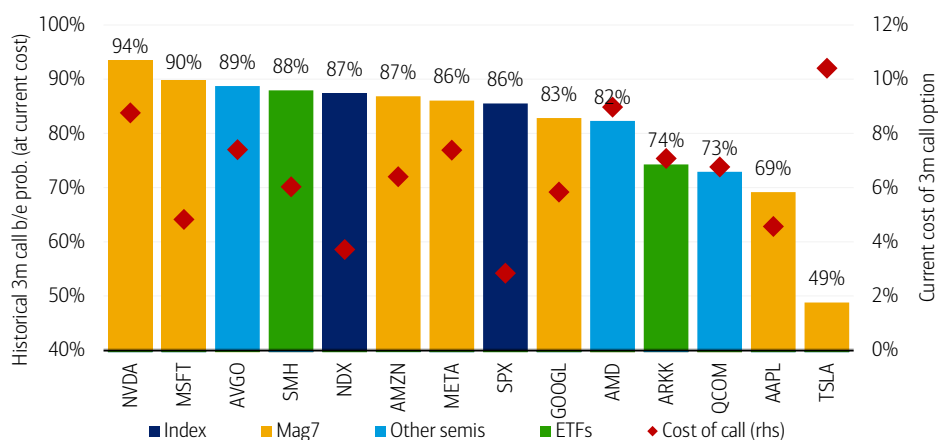
Given the high degree of uncertainty, optionality and vol-based solutions can offer safer conduits to hedge against the risks associated with bubbles. In particular, we find it safer to position for the upside using the asymmetry of options, even within the single stock space where the bulk of the AI focus has been. While prices for call options on these names may look high, historic breakeven rates at current prices are equally impressive (Exhibit 35), indicating that even the inverted call skew in many of these stocks (which could be accounting for serial correlation in returns) has been underpricing their momentum over the past year.

More systematically, options can present a better proposition than cash to hedge risk-off moves or buy dips, a concept we explore in our piece [Don't dash to cash; you have options](#) and in our Year Ahead report [Flying Blind](#).

Lastly, vol-based structures like owning up-variance or 'upvar', which accrues volatility when spot is above a certain threshold, can offer an efficient means to target the typical spot-up, vol-up dynamics of a bubble and hedge against its associated risks (see [06-Feb-24 GEVI](#) for more details).

Exhibit 35: Options today are underpricing recent equity momentum and can offer value in hedging upside risks. Call options on the S&P, Nasdaq and numerous popular AI and tech exposed equities would have broken even over 80% of the time at current pricing

3m ATM call option historical breakeven probability at current costs



Source: BofA Global Research. Data from 30-Nov-22 to 24-May-24 (introduction of ChatGPT to present). Indicative call pricing as of 24-May-24. Call "breakeven" represents periods when maximum return on initial spot over following 3m is greater than current cost of call option. "Magnificent 7" or Mag7 are defined as the group of Apple (AAPL), Amazon (AMZN), Alphabet (GOOGL), Meta (META), Microsoft (MSFT), Nvidia (NVDA) and Tesla (TSLA).

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Implications for the vol dispersion trader

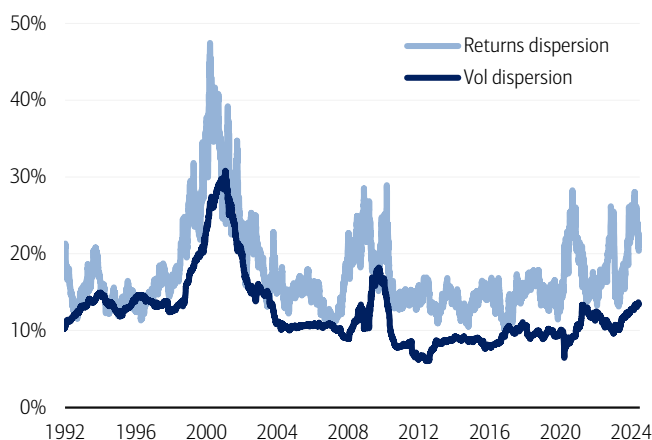
Relating vol dispersion to returns dispersion

Volatility dispersion, in this analysis, is measured as the spread between weighted average single stock volatility and index volatility. The measure can be calculated on an implied or realized basis, depending on the underlying volatility measures used.

Similar to how returns dispersion measures the average deviation of stock returns vs the overarching index, vol dispersion can be seen as a measure of the (weighted) average deviation of stock volatility vs index volatility. In fact, single stock volatility and correlation are the primary drivers of vol dispersion, with high single stock vol and low correlation contributing to higher vol dispersion (and vice versa). As a result, vol dispersion and returns dispersion track each other relatively well historically (though not exactly since they measure slightly different dynamics; see Exhibit 36 & Exhibit 37).

Exhibit 36: Just as returns dispersion rose sharply during the tech bubble, vol dispersion too reached historic levels that have not been seen since

S&P 500 12m returns dispersion & 12m realized vol dispersion

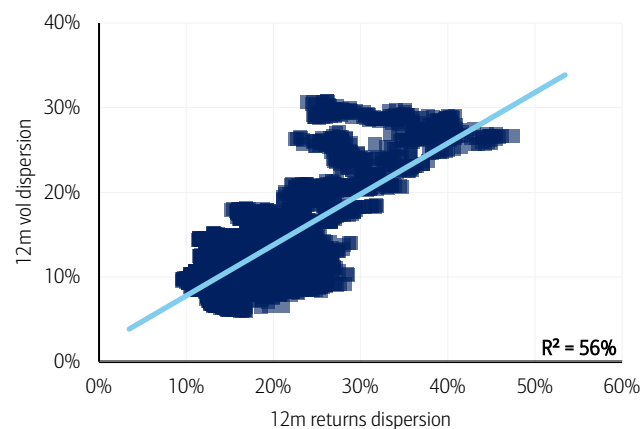


Source: BofA Global Research. Data from 1-Jan-92 to 31-May-24. Dispersion data calculated based on historical index membership and constituent weights (changing daily) for all member stocks of the S&P 500.

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Exhibit 37: While not capturing exactly the same dynamics in price action, returns dispersion and vol dispersion remain closely related measures driven by single stock vol and correlation

12m realized vol dispersion vs 12 returns dispersion for S&P 500



Source: BofA Global Research. Data from 30-Jan-91 to 31-May-24.

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A brief history of vol dispersion

Similar to returns dispersion, vol dispersion levels were historically high during the tech bubble, with single stock vol levels rising consistently and correlation falling aggressively towards the end of the bubble (Exhibit 38 & Exhibit 39). Looking back at the past decade and on an S&P Top 50 universe to proxy how the trade is conventionally implemented in practice, vol dispersion levels (both on an implied and realized basis) have trended higher, with a notable acceleration in the past two years. In fact, current implied levels are approaching those realized in the peak of the GFC in 2008.

Using the analogy of the tech bubble, the recent uptrend and acceleration in dispersion might suggest early indications of a bubble. However, current levels remain closer to those seen in 1995-96 than in 1999, in line with our previous observations. Moreover, much of the recent uptrend in dispersion has been driven by decorrelation rather than a rise in single stock vol (Exhibit 39). Consequently, in our view, despite levels looking high compared to more recent history, vol dispersion could have further room to run should the AI wave turn into bubble.

How much is left in the dispersion tank? Are there alternatives?

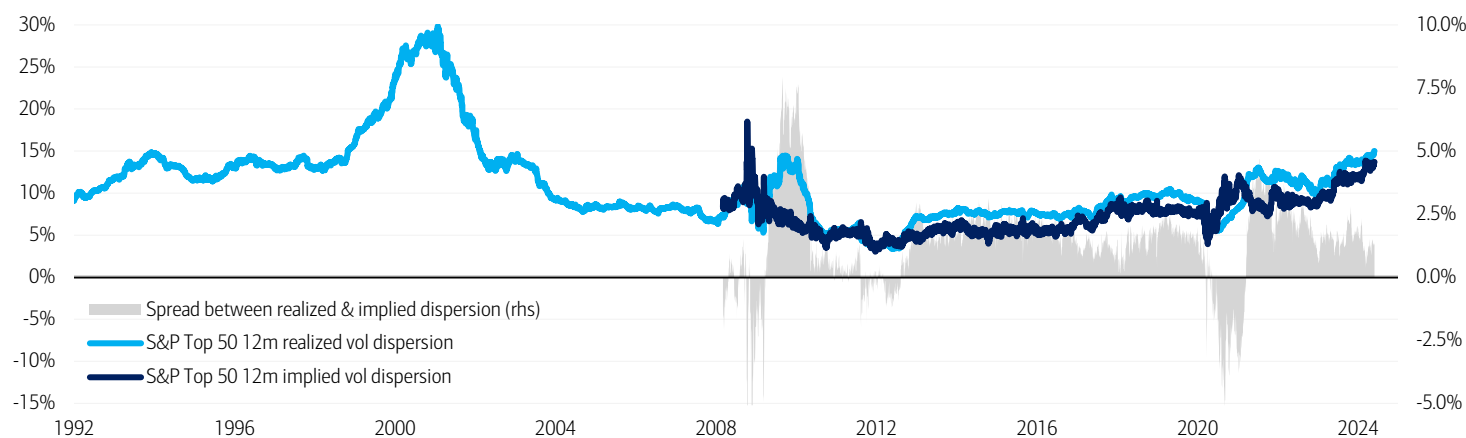
With implied dispersion levels slowly but steadily rising over the past decade or so, there is increasing concern that it's too late to enter vol dispersion trades, even if (trailing) realized vol dispersion has more than kept pace with implied over time (Exhibit 38). On

the one hand, a further rise in realized dispersion is not out of the question, particularly if we see more signs of bubble-like behavior in the form of rising single stock vols and low correlation/continued decorrelation. Indeed, although S&P Top 50 12m implied dispersion is currently in the 82nd percentile vs 12m realized dispersion since 1991, the right tail of the historical distribution is long courtesy of the late 90s, and realized levels could still double from here in an exact repeat of the tech bubble (Exhibit 38).

On the other hand, tail risks continue to loom given the risk of a correlated shock in today's uncertain environment, potentially crowded positioning in the dispersion space (though the exact size of the trade remains much debated), and unknowns around how multi-manager pod shops may manage exposure in a correlation spike. Investors particularly concerned about such risks may seek alternatives to "conventional" S&P Top 50 (vega-flat) vol dispersion, including, for example, more bespoke stock baskets, alternative funding legs, and/or other ways to cheapen outright long single stock vol exposure (as implied stock vol has lagged the rise in implied dispersion – Exhibit 40).

Exhibit 38: While dispersion levels have been rising since 2011, the scenario of an AI bubble similar to an internet bubble could create much further room to run in the trade

S&P Top 50 12m implied and realized vol dispersion

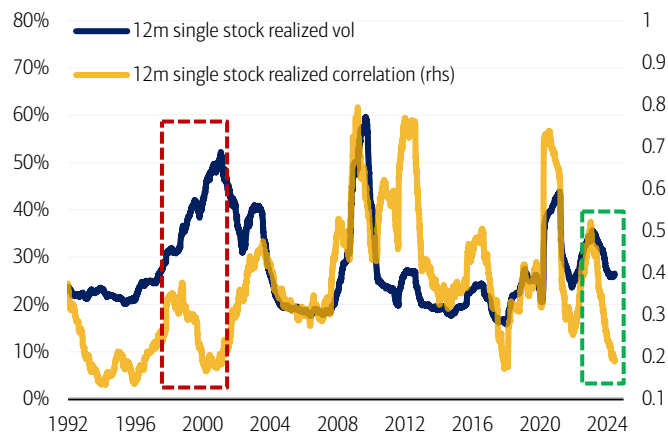


Source: BofA Global Research. Realized vol dispersion data from 1-Jan-92 to 24-May-24. Implied vol dispersion data from 3-Mar-08 to 24-May-24.

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Exhibit 39: The concurrence of rising stock vol and falling correlation drove vol dispersion in the tech bubble to historic highs; today's stocks are similarly decorrelated but less volatile

S&P Top 50 12m single stock realized vol and realized correlation

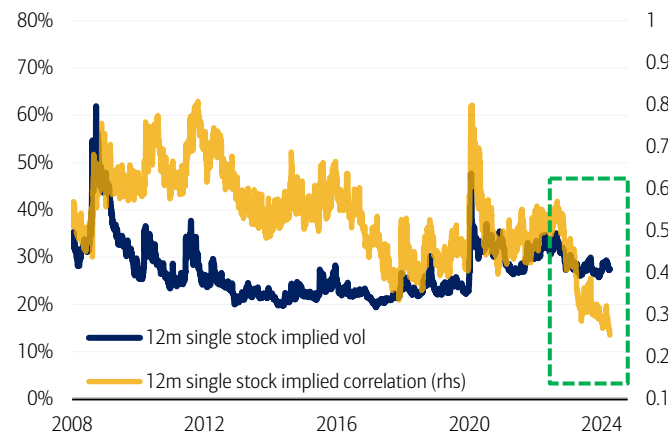


Source: BofA Global Research. Data from 1-Jan-92 to 24-May-24.

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Exhibit 40: Much of the rise in implied vol dispersion levels today has been driven by a sharp drop in implied correlation, with implied stock vol still lagging

S&P Top 50 12m single stock implied vol and implied correlation



Source: BofA Global Research. Data from 3-Mar-08 to 24-May-24.

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Potential Risk at Expiry & Options Limited Duration Risk

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